

QTM 220

Conditionally Randomized Experiments and Natural Experiments

December 4 and 6, 2018

Outline

Conditionally/Blocked Randomized Experiments

Conditional Example

Additive vs Interactive

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Conditional Example

Additive vs Interactive

(Marginally) Randomized Experiments

- Units/individuals are randomly assigned to treatment or control.
- Work great with large sample sizes and often work fine with medium sample sizes.
- With small/medium sample sizes, can get unlucky randomizations.
- In the blood pressure example, if we get unlucky we might assign treatment to young subjects and control to older subjects, hence treatment confounded by age.
- Can try to fix unlucky randomizations by controlling for age, but if treatment and age close to collinear, then VIF large.

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Conditionally/Blocked Randomized Experiments

- Units/individuals are randomly assigned to treatment or control within blocks (e.g., randomly assign among young, and randomly assign among old).
- Ensures some young and old assigned to both treatment and control.
- Can use different probability of treatment assignment in the block (e.g. 50-50 among young and 75-25 among old)
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Control Variables

In a marginally randomized experiment

- Simple regression of Y on T will be unbiased.
- Multiple regression of Y on T and X can be more efficient (especially in large samples).
- Bias/Variance tradeoff.

In a conditionally randomized experiment

- Simple regression of Y on T will be biased when probability of treatment assignment differs across blocks.
- Multiple regression of Y on T and X almost always more efficient.

Example: Conditionally Randomized Experiments

If treatments are randomly assigned conditionally on another variable, then the average $Y_i(1)$ among the $T_i = 1$ subjects **may not** equal the average $Y_i(1)$ among the $T_i = 0$ subjects.

i	T_i	Y_i	$Y_i(1)$	$Y_i(0)$	$Y_i(1) - Y_i(0)$
1	1	70	70	70	0
2	1	80	80	70	10
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What is the average $Y_i(1) - Y_i(0)$?

$$\begin{aligned}
 \frac{0 + 10 + 0 + 10 + 0 + 0}{6} &= \frac{20}{6} \\
 &\approx 3.3 \\
 &\neq 75 - 70
 \end{aligned}$$

Example: Conditionally Randomized Experiment

Suppose for one group ($X = 1$) a coin was flipped to assign treatment to half of the units, and for another group ($X = 0$), one out of four units was assigned to treatment.

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$$\frac{70 + 60 + 70 + 80}{4} = 70$$

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If we analyze each experiment separately, and average the results by the proportion in each experiment, then we get the ATE.

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$$\begin{aligned}
 & \frac{4}{6} \cdot \left\{ \frac{70}{1} - \frac{60 + 70 + 80}{3} \right\} + \frac{2}{6} \cdot \left\{ \frac{80}{1} - \frac{70}{1} \right\} \\
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 &= \frac{20}{6} = \frac{10 + 10}{6}
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Conditional Exchangeability

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Note this was also true for a (marginally) randomized experiment.

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Note that these would hold for a (marginally) randomized experiment.

Estimation with Conditionally Randomized Experiments

With a binary T and a binary X , we can always write the conditional expectation as a linear interactive function,

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$

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$$E[Y|T = 1, X = 1] = \beta_0 + \beta_1 + \beta_2 + \beta_3$$

Interpretation with Conditionally Randomized Experiments

Therefore, when T is randomized conditionally on X ,

$$\begin{aligned} E[Y|T = 1, X = 0] - E[Y|T = 0, X = 0] &= (\beta_0 + \beta_1) - \beta_0 \\ &= \beta_1 \end{aligned}$$

can be interpreted as the ATE among the $X = 0$ subjects.

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can be interpreted as the ATE among the $X = 1$ subjects.

$$\begin{aligned} E[\beta_1 + \beta_3 \cdot X] &= \beta_1 + \beta_3 \cdot E[X] \\ &= (1 - E[X])\beta_1 + E[X](\beta_1 + \beta_3) \end{aligned}$$

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can be interpreted as the ATE

Statistical Inference with Randomized Experiments

If $\beta_3 = 0$, and the four descriptive assumptions hold, then we can get approximate CIs and tests for the ATE by running the additive regression of Y on T and X .

If $\beta_3 \neq 0$, and the four descriptive assumptions hold then we can get approximate CIs and tests for the ATE by subtracting the mean $E[X]$ off of all X values, running the interactive regression of Y on T , $X - E[X]$, and $T \cdot (X - E[X])$ and using the estimate and standard error associated with coefficient on T .

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Outline

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Conditional Example

Additive vs Interactive

Washington (2008) example

Washington (2008) examines the effect of having daughters on the AAUW scores for members of congress (MCs) by comparing MCs with more daughters to MCs with fewer daughters, conditional on family size.

Omitted Variable Bias

Suppose we believe that a two variable linear model is the true model in the Washington (2008) example,

$$Y_i = \beta_0 + \beta_1 \text{Girls}_i + \beta_2 \text{Children}_i + \epsilon_i$$

but suppose that we omit the *Children* variable from the analysis

$$Y_i = \beta_0^* + \beta_1^* \text{Girls}_i + \epsilon_i^*$$

If we write the model for children as the following:

$$\textit{Children}_i = \gamma_0 + \gamma_1 \textit{Girls}_i + \nu_i$$

then the two explanatory variable model can be re-written as the following:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 \textit{Girls}_i + \beta_2 \textit{Children}_i + \epsilon_i \\ &= \beta_0 + \beta_1 \textit{Girls}_i + \beta_2(\gamma_0 + \gamma_1 \textit{Girls}_i + \nu_i) + \epsilon_i \\ &= \underbrace{\beta_0 + \beta_2\gamma_0}_{\beta_0^*} + \underbrace{(\beta_1 + \beta_2\gamma_1)}_{\beta_1^*} \textit{Girls}_i + \underbrace{\epsilon_i + \beta_2\nu_i}_{\epsilon_i^*} \end{aligned}$$

Other Conditional Models

What are some other ways to model the Washington (2008) example?

$$Y_i = \beta_0 + \beta_1 \mathbf{Girls}_i + \beta_2 \mathbf{Children}_i + \epsilon_i$$

Equation 1 from Washington (2008) re-written:

$$\begin{aligned} Y_i = & \beta_0 + \beta_1 \mathbf{Girls}_i + \beta_2 \mathbf{1}_{\{\mathbf{Children}_i=1\}} \\ & + \beta_3 \mathbf{1}_{\{\mathbf{Children}_i=2\}} \\ & + \vdots \\ & + \beta_6 \mathbf{1}_{\{\mathbf{Children}_i=5\}} + \epsilon_i \end{aligned}$$

Other Conditional Models

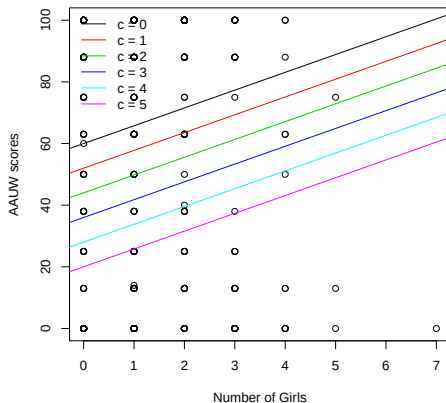
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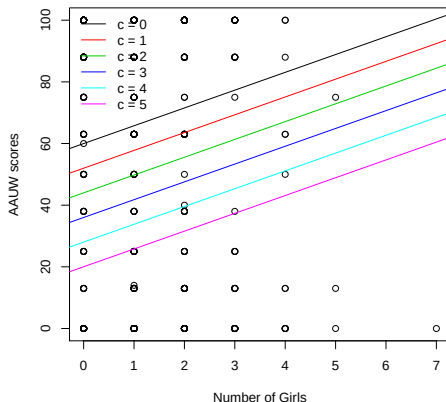
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The Additive Linear Model (105th Congress)



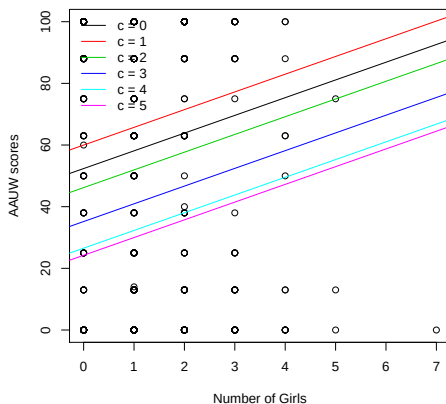
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The Additive Linear Model (105th Congress)

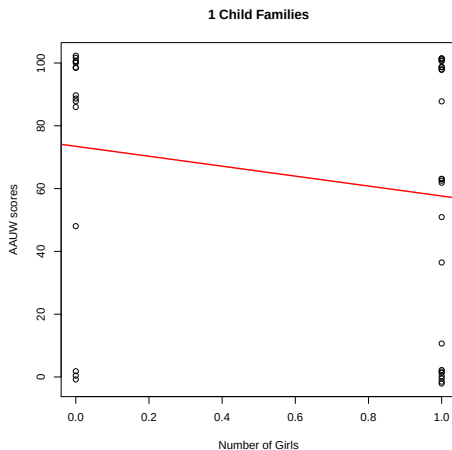


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Washington (2008) Eq. 1 Model

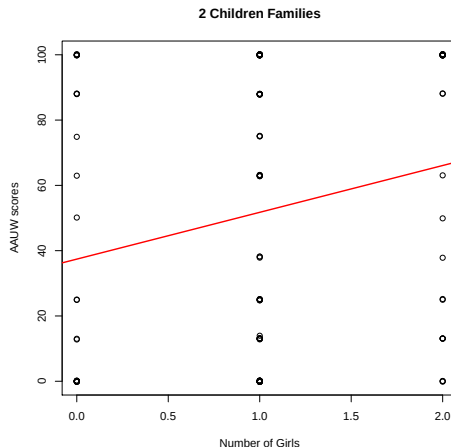


One Child Legislators in the 105th Congress



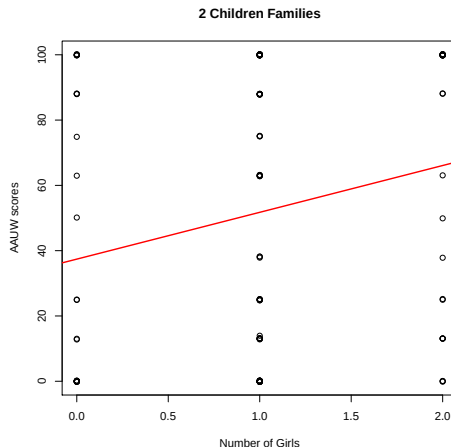
Among the subset of one-child legislators, we could adopt the standard potential outcomes approach.

Two Children Legislators in the 105th Congress



How would you construct the potential outcomes table for families with 2 children? What assumption is now implicit in the regression?

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This linear approach would also be straightforward for 3 children, 4 children, etc.

What are some ways we could combine these analyses across all legislators?

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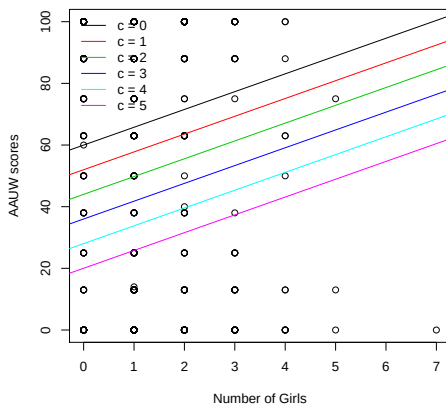
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The Additive Linear Model



The Interactive Linear Model

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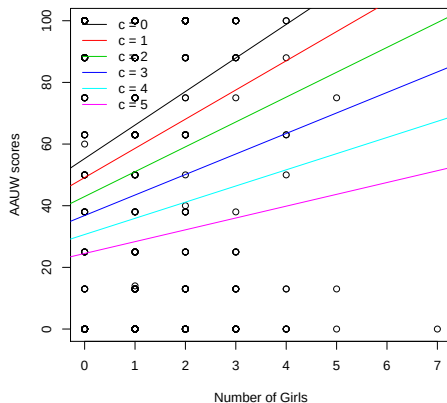
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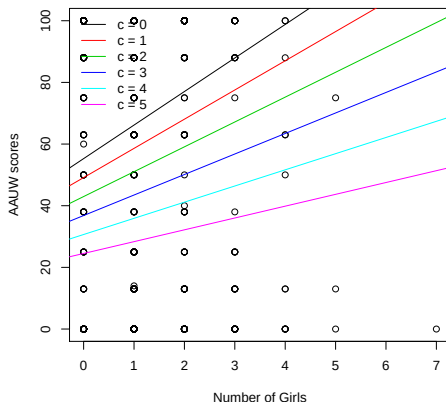
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Standardization in the Interactive Linear Model

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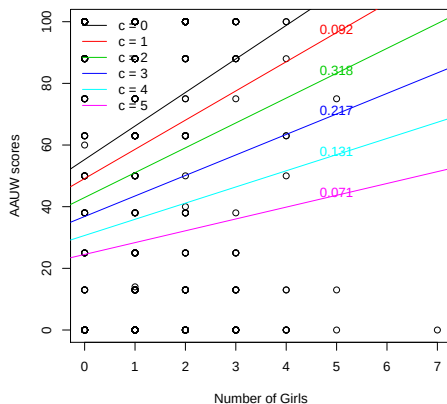
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Standardization in the Interactive Linear Model



Outline

Conditionally/Blocked Randomized Experiments

Conditional Example

Additive vs Interactive

Simple Regression

Don't use simple regression with a conditionally randomized experiment or a conditionally randomized natural experiment!

Leads to omitted variable bias.

Additive vs Interactive (Binary T and Binary X)

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x$$

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x + \beta_3 \cdot t \cdot x$$

- Interactive requires standardization ($P(X = 1)CATE_{X=1} + P(X = 0)CATE_{X=0}$) but provides an estimate of ATE.
- Additive is easier (just report β_1), but it provides an estimate of a weighted ATE.

```
cong12 <- subset(cong, totchi==1 | totchi==2)
cong12$anygirls <- cong12$ngirls > 0
cong12$twochi <- cong12$totchi==2
```

```
> lm(aauw ~ anygirls * I(twochi - mean(twochi)), da
(Intercept)    anygirlsTRUE      I(twochi - mean(twoch
      47.742           7.893                -
anygirlsTRUE:I(twochi - mean(twochi))
                        30.595
```

```
> lm(aauw ~ anygirls + twochi, data=cong12)
(Intercept)    anygirlsTRUE      twochiTRUE
      60.001           5.719        -12.501
```

ATE

```
> lm(aauw ~ anygirls * I(twochi - mean(twochi)), da
(Intercept)      anygirlsTRUE      I(twochi - mean(twoch
      47.742              7.893              -
anygirlsTRUE:I(twochi - mean(twochi))
                        30.595
```

```
mod <- lm(aauw ~ anygirls * twochi ,data=congl2)
```

```
p0 <- (1-mean(congl2$twochi))
```

```
p1 <- mean(congl2$twochi)
```

```
mod$coef[2]*p0 +
sum(mod$coef[c(2,4)])*p1
      7.893438
```

Weighted ATE

```
> lm(aauw ~ anygirls + twochi, data=cong12)
(Intercept)  anygirlsTRUE  twochiTRUE
      60.001         5.719      -12.501
```

```
mod <- lm(aauw ~ anygirls * twochi ,data=cong12)
```

```
w0 <- p0*mean(cong12$anygirls[cong12$twochi==0]) *
(1-mean(cong12$anygirls[cong12$twochi==0]))
w1 <- p1*mean(cong12$anygirls[cong12$twochi==1]) *
(1-mean(cong12$anygirls[cong12$twochi==1]))
```

```
mod$coef[2] * (w0 / (w0+w1)) +
sum(mod$coef[c(2,4)]) * (w1 / (w0+w1))
      5.71872
```

Additive vs Interactive (Binary T and general $X_1, X_2, \dots$)

With binary T and general $X_1, X_2, \dots$

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x_1 + \beta_3 \cdot x_2 + \dots$$

$$E[Y|T = 1, X = x] = f_1(x_1, x_2, \dots)$$

$$E[Y|T = 0, X = x] = f_0(x_1, x_2, \dots)$$

$$E_X[f_1(x_1, x_2, \dots) - f_0(x_1, x_2, \dots)]$$

- Interactive requires standardization (fit a model to the treated units, fit a model to the control units, calculate differences between them for all observations).
- Additive is easier (just report β_1), but it provides an estimate of a weighted ATE.

Additive vs Interactive (Non-Binary T and general $X_1, X_2, \dots$)

With non-binary T and general $X_1, X_2, \dots$

$$E[Y|T = t, X = x] = \beta_0 + \beta_1 \cdot t + \beta_2 \cdot x_1 + \beta_3 \cdot x_2 + \dots$$

- Interactive requires ... too hard. Come see me if you really want to try.
- Additive is easier (just report β_1), but it provides an estimate of a weighted-weighted ATE.